

Multiple Choice (10 marks)

1. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
 - (a) Recursive procedures
 - (b) Arbitrary goto's
 - (c) Two-dimensional arrays
 - (d) Integer-valued functions
2. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
3. To verify a digital signature we need the
 - (a) Sender's Private key
 - (b) Sender's Public key
 - (c) Receiver's Private key
 - (d) Receiver's Public key
4. Which of the following is not a block cipher operating mode?
 - (a) ECB
 - (b) CFB
 - (c) CBF
 - (d) CBC
5. Which of the following characteristics of a programming language is best specified using a context-free grammar?
 - (a) Identifier length
 - (b) Maximum level of nesting
 - (c) Operator precedence
 - (d) Type compatibility

True / False (10 marks)

Answer True or False

6. True or False: An Abstract Syntax Tree (AST) is responsible for managing information about variables and their attributes in a compiler.
7. True or False: A secure pseudorandom permutation (PRP) can be constructed from a secure pseudorandom generator (PRG) using the GGM construction and Luby-Rackoff theorem.
8. True or False: The Silk Road was a prominent Dark Web marketplace known for the sale of illegal drugs and various illicit goods.
9. True or False: Only biometric data is considered a valid authentication method in modern security practices.
10. True or False: The exploitation of the Heartbleed bug allows for reading data outside the bounds of a buffer in memory.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check whether the given ip address is valid or not using regex.
12. Write a python function to find minimum sum of factors of a given number.

End of Paper